Pre-work · Muscle and the Muscles of the Trunk and Upper Limb

Muscle Structure and Sarcomeres, and Fascicle Arrangement and Lever Systems. Read both Part A chapters first. The last two visuals and the last drawing are lab prep: the named muscles are anatomy lab material, so pull their origin, insertion, and action from the muscle chart on the Loops page, not from Part A.

Module 3 · 1 of 5 | Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read** Read the Part A chapter first. This packet does not repeat it. It puts it to work.
2. **Organize** Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
3. **Draw** Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply** Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve** Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the two muscle chapters into mini visuals, then use the last two to get ahead of lab. Eight chunks. The grids carry the naming, the hubs carry the structure.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A muscle has short fascicles set at an angle on both sides of a central tendon. Predict its power and its range of motion, and name a muscle built this way.

Answer. High power, small range of motion. Bipennate. Rectus femoris is an example.

Reasoning. Start with the geometry, not the name. Angling the fascicles lets far more of them pack into the same volume of muscle, and every one of them pulls on the tendon, so force adds up. But each fascicle is short, and a fiber can only shorten to about two thirds of its resting length, so the tendon does not travel far. Power and range trade against each other, and the architecture is where the trade is made visible.

takes longer than three minutes you are copying instead of organizing.

1 LADDER

Levels of muscle organization

Five rungs, muscle at the top down to myofilament at the bottom. Write the wrapping beside each rung that has one. Three rungs get a sheath and two do not, and that is the point.

2 HUB

The sarcomere

One sarcomere across the middle, Z disc to Z disc. Spoke out to A band, I band, H zone, M line, and zone of overlap. Shade which filaments sit in each. This is the most tested figure in the chapter.

3 SPLIT

Thick against thin filament

Main protein, structure, and what it anchors to. Hang tropomyosin and troponin off the thin branch, titin and myomesin off the thick.

4 GRID

The three muscle tissues

Skeletal, cardiac, smooth down the side. Location, striations, control, and cell features across the top. Star the two features that are unique to one tissue each.

5 GRID

Fascicle arrangement

Pattern down the side, arrangement, example muscle, and power against range across the top. Six rows. The last column is the one that answers application questions.

DRAWING 1

A sarcomere, fully labelled

BRAIN DUMP FIRST

Two minutes. Every band, line, and zone, and which filament sits where.

Draw one sarcomere large, Z disc to Z disc. Lay in the thick filaments anchored at the M line and the thin filaments anchored at both Z discs. Now shade and label the A band, both I bands, the H zone, the M line, and the zone of overlap. Add titin running Z disc to M line.

In your notebook, head the page **M3–1 A sarcomere, fully labelled** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can say which regions narrow when the muscle shortens, and which one never changes, and point to why on your own drawing.

Set A · Name it

1. The sheath wrapping a single muscle fiber is the ______________________.
2. The structural unit of one T tubule flanked by two terminal cisternae is a ______________________.
3. The band that contains thin filaments only is the ______________________.
4. The elastic protein running from the Z disc to the M line is ______________________.
5. Name the three lever classes and one body example of each.

Set B · Predict it

1. A muscle is rebuilt from parallel to pennate architecture. Predict what happens to its power and to its range.
2. A tendon is torn from the bone. Predict how the pull of the individual fibers is affected, using the three sheaths.
3. Dystrophin is faulty in a muscle fiber. Predict what happens with ordinary use, and say why.
4. A tendon is moved surgically so it inserts farther from the joint. Predict the effect on strength and on speed.

Set C · Tell it apart

1. Distinguish a muscle strain from a tendon rupture by which structure fails and how each heals.

6 **LADDER**

The three lever classes

Three rungs, first through third. Draw the bar with fulcrum, effort, and load in position on each rung, then write the body example beside it.

7 **GRID**

Superficial back and anterior thorax, lab prep

Muscle down the side, origin, insertion, action, and innervation across the top. Nine rows, filled from the muscle chart on the Loops page. Then add one more column: which scapular movement each produces, and circle the antagonist pairs.

8 **HUB**

The rotator cuff and the shoulder, lab prep

Head of the humerus at the hub. Spoke to supraspinatus, infraspinatus, teres minor, subscapularis, and mark the tubercle each reaches. Add deltoid, teres major, and coracobrachialis around the outside. Origin, insertion, and action come from the muscle chart.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

A muscle in cross section, from sheath to fiber

BRAIN DUMP FIRST

One minute. The three sheaths, what each wraps, and how they reach the tendon.

Draw a muscle cut across. Outline the whole muscle with the epimysium, divide the inside into fascicles each ringed by perimysium, and draw a handful of fibers inside one fascicle each ringed by endomysium. Then draw the muscle from the side and show all three sheaths merging into the tendon at the end.

In your notebook, head the page **M3-2 A muscle in cross section, from sheath to fiber** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can trace a line with your finger from inside one fiber all the way to the bone without lifting it off the page.

2. Compare the three muscle tissues by striations and control, and name the one feature that places each on a slide.

3. Explain how the banding of a sarcomere produces the striations you see down a microscope.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

The posterior shoulder and back, in layers, lab prep

BRAIN DUMP FIRST

Two minutes. Every superficial back muscle and the rotator cuff underneath.

Draw the back of the trunk and scapula. Lay in trapezius and latissimus dorsi as the superficial layer on the left half. On the right half draw them peeled away to show levator scapulae, both rhomboids, and the rotator cuff on the scapula. Mark the origin and insertion of each with a dot and a line.

In your notebook, head the page **M3-3 The posterior shoulder and back, in layers, lab prep** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can name what happens to the scapula when each muscle you drew contracts, without reading the label.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 Levels of muscle organization

LADDER

2 The sarcomere

HUB

3 Thick against thin filament

SPLIT

4 The three muscle tissues

GRID

5 Fascicle arrangement

GRID

6 The three lever classes

LADDER

7 Superficial back and anterior thorax, lab prep

GRID

8 The rotator cuff and the shoulder, lab prep

HUB